An Exact Fold Condition for Mass-Balanced Models of Protein Quality Control

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Abstract

Supplementary material accompanying the paper above. Every number in the caption is recomputed by the script named in it and asserted by the accompanying test suite.

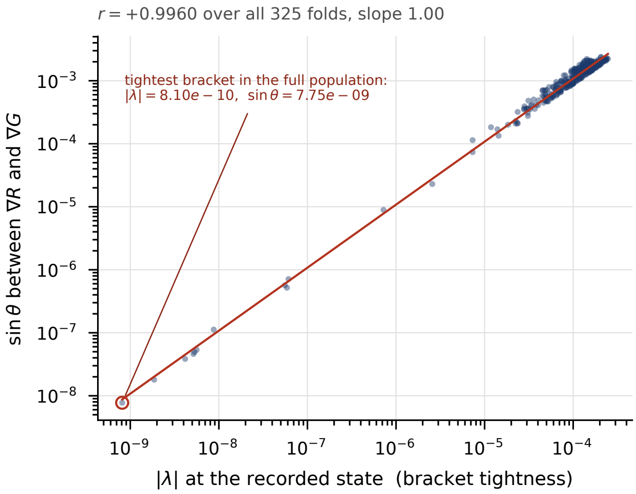

Fig. S1 The parallelism residual is bracket tolerance, not failure of the identity. Over all 325 folds of the load grid, $\sin\theta$ between ∇R and ∇G tracks the leading eigenvalue recorded at the same state across five decades, with a log–log correlation of $+0.9960$ and a slope of 1.00 — the residual is *proportional* to how loosely the state is bracketed, which is what bracket tolerance means and what a failing identity would not produce. The circled point is the tightest bracket in the population, $|\lambda| = 8.10 \times 10^{-10}$ with $\sin\theta = 7.75 \times 10^{-9}$. Normalisation, reported here rather than given a panel of its own: on these same 325 states the max-normalised residual correlates with $|\lambda|$ at -0.835 and reaches 1.54×10^{-2}, while the gradient-normalised residual used throughout correlates at $+0.041$ and reaches 1.29×10^{-9}. The first degrades as the bracket tightens on the object it is meant to verify, because both $\det J$ and the cross product vanish at a saddle-node and their ratio is 0/0 there; the second does not. Every number in this caption is recomputed by `scripts/figures/fig_identity.py:captionNumbers` and asserted by the test suite.